01 / PROBLEM AND CONTRIBUTION

# 32 nm four-bit CMOS adder

EEE 591 · Digital System Circuits | Fall 2025 | Individual design

**RESULT AT A GLANCE**

Implemented a four-bit adder occupying 40.083 μm², with 277 ps extracted worst-case delay and 47.3 μW average power; compared it with the 152 ps schematic result.

## The engineering question

How does a four-bit CMOS adder's timing change when layout parasitics are included?

## Situation

A fast schematic is only part of a custom digital circuit: routing and device parasitics affect the extracted result. This project compares both views of a 32 nm, four-bit adder.

## Task

Size the critical carry path, implement the layout, run DRC/LVS, and characterize delay and power after extraction.

## My contribution

Transistor sizing, layout, physical verification, and extracted simulation.

## Tools and concepts

CMOS / IC layout / DRC / LVS / Transistor sizing / Post-layout simulation

02 / DESIGN AND METHOD

# From the task to implementation

### 01 / Implementation step

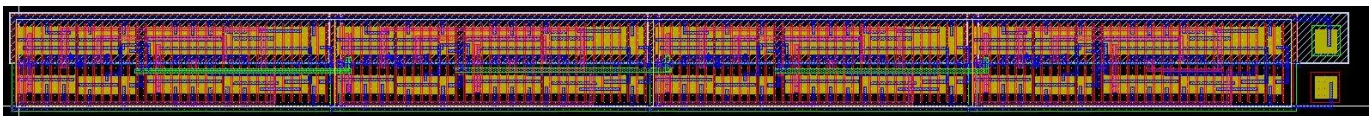
Sized carry-path and output-stage transistors for drive strength while retaining minimum-size devices where the design allowed.

### 02 / Implementation step

Implemented the four-bit layout, ran DRC and LVS, and compared schematic-level and extracted simulations.

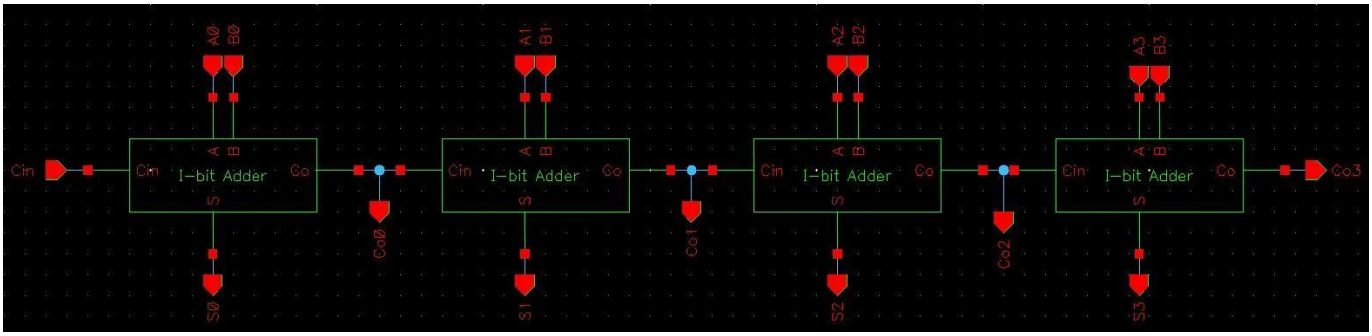
### Critical-path sizing

The implementation emphasizes the carry path and output drive. Comparing schematic and extracted behavior makes the influence of physical interconnect and device parasitics visible in the same design task.



**Four-bit adder layout**

Custom layout from the 32 nm CMOS adder report.




**Four-bit adder schematic**

Schematic hierarchy used for the custom adder.

03 / RESULTS AND ENGINEERING PERSPECTIVE

# Evidence, interpretation, and next steps

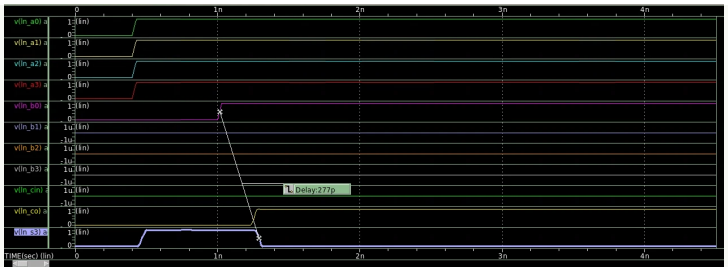
| | |
|---|---|
| **Post-layout delay** | 277 ps |
| **Average power** | 47.3 µW |
| **Layout area** | 40.083 µm² |

## Reported result

The report records a worst-case delay of 152 ps before layout and 277 ps after extraction, with 47.3 µW average power and a layout area of 40.083 µm².

## Using the extracted result

The extracted delay is 277 ps compared with 152 ps before layout. Area and average power are reported alongside delay to describe the implementation more completely; none of these figures establishes a silicon measurement.




**Extracted timing waveform**

Post-layout timing simulation from the report.

## Scope and limitations

The post-layout result is the relevant timing figure for this case study. The comparison makes the impact of physical implementation visible instead of presenting only the faster schematic simulation.

### Possible next steps - proposed, not completed results

- Add the test-vector and loading conditions next to the timing/power table in the source repository.
- Explore carry-path/layout changes with a consistent post-extraction comparison.

Source material: EEE 591 Digital System Circuits Lab 4 report, pp. 1-4; approved adder brief.

Prepared from the supplied reports, project files, and verified portfolio copy. Figures retain their original content; explanatory diagrams are labelled.